

FORM: DISCLOSURE STATEMENT

The provision of the data is subject to the User providing the South African Weather Service (SAWS) with a detailed and complete disclosure, in writing and in line with the requirements of clauses 1.1 to 2.4 (below), of the purpose for which the specified data is to be used.

- 1 **Should the User intend using the specified data for commercial gain then the disclosure should include the following:**
 - 1.1 the commercial nature of the project/funded research project in connection with which the User intends to use the specified data;
 - 1.2 the names and fields of expertise of any participants in the project/funded research project for which the specified data is intended; and
 - 1.3 the projected commercial gains to the User as a result of the intended use of the specified data for the project/funded research project.
- 2 **Should the User intend using the specified data for the purposes of conducting research, then the disclosure should include the following:**
 - 2.1 the title of the research paper or project for which the specified data is to be used;
 - 2.2 the details of the institution and supervisory body or person(s) under the auspices of which the research is to be undertaken;
 - 2.3 an undertaking to supply SAWS with a copy of the final results of the research in printed and/or electronic format; and
 - 2.4 the assurance that no commercial gain will be received from the outcome from the research.

If the specified data is used in research with disclosure being provided in accordance with paragraph 2 and the User is given the opportunity to receive financial benefit from the research following the publication of the results, then additional disclosure in terms of paragraph 1 is required.

The condition of this disclosure statement is applicable to the purpose and data requirements of the transaction recorded in Schedule 1 below.

SCHEDULE 1

Please note: The South African Weather Service will only act upon customer requirements noted on this disclosure statement and not from any other correspondence.

FULL PERSONAL DETAILS OF USER

Full Names	Dr Antje Steinfurth ¹ /Dr Emily G. Simmonds ²
University/school/organisation	¹ RSPB Centre for Conservation Science, David Attenborough Building, Pembroke Street, Cambridge, CB2 3QZ, UK ² Institute of Ecology and Evolution, School of Biological Sciences, University of Edinburgh, Scotland, UK
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Supervisor	Dr Antje Steinfurth Dr Emily G. Simmonds

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Project/Thesis Title	Identifying a signal of climate drivers on the breeding success of Southern Giant Petrels (<i>Macronectes giganteus</i>) on Gough Island
Current registered degree (e.g. BSc)	MSc
Expected finalization date (MMYYYY)	31/12/2026

The South African Weather Service reserves the right to request, at any time, from the student proof of registration for the Degree at the University.

THE PURPOSE (Please indicate a detailed description of the purpose for which the data will be used)

Aims and background:

Gough Island in the central South Atlantic, part of the UK Overseas Territory of St Helena, Ascension and Tristan da Cunha, is considered one of the most important seabird breeding sites globally. Despite its small size (~65 km²), the island supports an estimated eight million birds of at least 24 species, many of which are endemic or near-endemic to the island or island group of Tristan da Cunha. The primary threat to the island's seabird populations is invasive, non-native House Mice (*Mus musculus*). Introduced in the 19th century, the mice have caused substantial declines in breeding success in many species breeding on the island, particularly those that nest below-ground or breeding during winter. A mouse eradication attempt was carried out in the austral winter of 2021, but the operation ultimately proved unsuccessful, and invasive mice remain on the island. However, mouse predation is not the only threat to these birds. Other pressures, such as, global climate change and fisheries, are also likely influencing breeding success, although the extent of these impacts remain poorly understood.

This project aims to explore how climate drivers (through weather related changes, such as wind speed, rainfall, temperature) may be impacting breeding success of Gough Island's seabirds. Unlike many other seabirds on Gough, there is currently no evidence that Southern Giant Petrels (*Macronectes giganteus*) are affected by mouse predation, making them the ideal species in which to investigate environmental drivers of breeding success. If this approach proves successful, it will be extended to species along a gradient of increasingly mouse-impacted populations, with the Atlantic Yellow-Nosed Albatross (*Thalassarche chlororhynchos*) as the second candidate species but ultimately expanding across all species breeding on Gough Island. This study will apply demographic and population modelling techniques to estimate current breeding success, identify potential climate drivers and their effects, and potentially forecast future population trajectories. By revealing previously unknown factors influencing breeding success, the project will address a critical knowledge gap and may help guide conservation strategies on the island, including strengthening the case for a future second mouse eradication attempt.

Questions/aims:

- What are the current estimates of breeding success for the Southern Giant Petrels and Atlantic Yellow-nosed Albatross on Gough Island. and how do these compare with populations (of similar species) on predator free islands?
- Can we identify a signal of climate drivers influencing breeding success in Southern Giant Petrels?
- Can we identify a signal of climate drivers influencing breeding success in Atlantic Yellow-Nosed Albatross?
- What do these findings imply for future management strategies on Gough Island?

The objectives of this research are to assess: (1) current breeding success for the Southern Giant Petrel and Yellow-nosed Albatross compared with populations on predator free islands, (2) significant climatic drivers influencing breeding success of the Southern Giant Petrel and Atlantic Yellow-nosed Albatross, and, (3) significant climatic drivers influencing breeding success among the three sub-colonies of Southern Giant Petrels. Application of these aims for

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the secondary species, the Atlantic Yellow-nosed Albatross is time contingent on the success of the study of the Southern Atlantic Petrel but have been optimistically included should time permit.

Proposed method:

This study will use RSPB's long-term monitoring database on breeding success and numbers of breeding pairs, combined with meteorological data from the Gough Island Weather Station, to:

- Quantify reproductive processes for the study species – using regression-based analyses (GLMMs)
- Fit covariates into the models to cover environmental impacts (e.g. temperature, precipitation, storms) to determine environmental drivers of reproductive success.
- Apply the same method on gradually more mouse-impacted species to determine the relative impact of weather on species under multiple pressures.
- There will be options to explore Bayesian methods for this project, but it is not essential.

Expected outcomes:

This project will provide an important insight into potential climate drivers of breeding success in Gough Island's seabirds. This will contribute to Tristan da Cunha's management plan for the island and provide valuable understanding of how breeding success may respond once the major threat of mice is removed. This work could also form the basis of a scientific publication (we will aim for this).

DATA REQUIRED *(Indicate weather elements (e.g. rain, temp), place/s, time period and resolution (e.g. daily, hourly))*

Weather elements: Rainfall, temperature, and wind speed/direction (if the latter is available – rain and temperature are the key).

Resolution: Daily mean of each weather element.

Time period: Time period is July to May from 2000-2025.

Place: Gough Island, Weather Station

I hereby accept that:

- SAWS will be acknowledged in the resulting thesis/project or when published, for the data it provided.
- SAWS will be provided with a copy of the final results in printed or electronic format.
- The data received shall not be provided to any third party.

Signature of the User:

Date:

(Please sign the document and do not type your name in as this is a legal document and requires a signature.)

PROTECTION OF PERSONAL INFORMATION ACT (POPIA)

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SAWS value its customer's privacy and strives to continuously ensure compliance with POPIA, to protect and safeguard against unauthorised use of personal information.

The Customer hereby accepts that:

- He/she has read and understood the customer processing POPIA notice found on <https://www.weathersa.co.za/home/popia>

Signature of the User:

Date:

(Please sign the document and do not type your name in as this is a legal document and requires a signature.)